

Completing the level-3 and level-4 frontier: the $w = 1$ mixed cube-root double polylogarithms

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Companion to *Multiple polylogarithms at Gaussian and Eisenstein arguments*
(gaussian-eisenstein-double-polylogs).

Abstract

The article *Multiple polylogarithms at Gaussian and Eisenstein arguments* left one piece unfinished: the *mixed* cube-root points $\text{Li}_{a,b}(\rho^2, \rho)$ (and the Gaussian $\text{Li}_{a,b}(i, -i)$), where $w = z_1 z_2 = 1$ degenerates the accelerated evaluator. This report builds a robust $w = 1$ evaluator—a smooth-correction identity accelerated by Wynn’s ρ -algorithm—generates the previously inaccessible values to 500 digits, reduces them to explicit closed forms through weight 4, and completes the full level-3 and level-4 depth-2 double-shuffle dimension. The headline: in *both* towers the mixed $w = 1$ point adds, beyond the article’s survivors, exactly *two* new depth-2 constants—a real $\text{Re Li}_{4,1}$ direction at weight 5 and an imaginary $\text{Im Li}_{5,1}$ direction at weight 6—in identical positions.

1 The one remaining piece

The companion article closed its level-3 (Eisenstein) and level-4 (Gaussian) double-shuffle dimension computations with one explicitly flagged gap. In the Eisenstein run the *mixed* cube-root points

$$\text{Li}_{a,b}(\rho^2, \rho), \quad \text{Li}_{a,b}(\rho, \rho^2), \quad \rho = e^{2\pi i/3},$$

were never folded in, because there $w = z_1 z_2 = \rho^3 = 1$, and (quoting the article)

both the Lerch-plus-Wynn evaluator and a direct cumulative-sum evaluator stall there at ~ 8 digits—the period-three, non-oscillating tail defeats Wynn’s ε -algorithm ... Folding the mixed cube-root points in (for the full level-3 frontier) requires a period-three-aware accelerated evaluator, or Wolfram; it is the one remaining piece.

The Gaussian run skipped its own $w = 1$ point $(i, -i)$ for the same reason. This report builds the missing evaluator, generates the previously inaccessible values to several hundred digits, and uses them to (i) harvest closed forms for the mixed-point values and (ii) complete the full level-3 and level-4 depth-2 double-shuffle dimension.

Throughout, the convention is

$$\text{Li}_{a,b}(z_1, z_2) = \sum_{n_1 > n_2 \geq 1} \frac{z_1^{n_1} z_2^{n_2}}{n_1^a n_2^b}, \quad (1)$$

which is `MultiplePolyLog[{a,b},{z1,z2}]` in Wolfram 15 (exponents first).

2 Why the standard evaluator degenerates at $w = 1$

With the convention (1), the accelerated evaluator of the article writes the inner tail as a Lerch transcendent and sums the *outer* series with an acceleration factor $w^n = (z_1 z_2)^n$. At the mixed points $w = 1$, that factor is identically 1: the oscillation Wynn’s ε -algorithm relies on disappears, leaving a bare period-three power-law tail that the algorithm cannot resolve past a handful of digits. Summing over the *larger* index instead keeps an oscillating factor $z_1^{n_1}$, but for $z_1 = \rho^2$ this is a *period-three* oscillation, and a plain ε -algorithm (tuned for geometric tails) again stalls.

3 A robust $w = 1$ evaluator

Two ingredients fix it.

(a) A smooth correction identity. Writing $l = n_1$ (the larger index) and splitting the inner partial sum $P(l) = \sum_{n < l} z_2^n / n^b = \text{Li}_b(z_2) - R(l)$ with tail $R(l) = \sum_{n \geq l} z_2^n / n^b = z_2^l \Phi(z_2, b, l)$ (Lerch Φ), one gets the exact identity

$$\boxed{\text{Li}_{a,b}(z_1, z_2) = \text{Li}_a(z_1) \text{Li}_b(z_2) - \sum_{l \geq 1} \frac{w^l}{l^a} \Phi(z_2, b, l)} \quad (2)$$

At $w = 1$ the correction summand $\Phi(z_2, b, l)/l^a$ is *smooth*—every oscillating factor $z_1^l z_2^l = w^l$ collapses to 1—with the clean asymptotic $\Phi(z_2, b, l) \sim \sum_{j \geq 0} \binom{-b}{j} c_j(z_2) l^{-b-j}$, where $c_0 = \frac{1}{1-z_2}$ and $c_{j \geq 1} = \text{Li}_{-j}(z_2)$ (rational in z_2). The subtle point—the source of a factor-of-one bug that cost five digits in a first draft—is that $c_0 = \sum_{k \geq 0} z_2^k = \frac{1}{1-z_2}$ is *not* $\text{Li}_0(z_2) = \frac{z_2}{1-z_2}$; they differ by exactly 1. This gives an Euler–Maclaurin tail in Hurwitz zetas, $\sum_{l > M} \Phi/l^a = \sum_j \binom{-b}{j} c_j \zeta(a+b+j, M+1)$, a fast and accurate cross-check at low to medium precision.

(b) Period-aware ρ -acceleration for production. For high precision the production evaluator sums the outer cumulative series directly, samples the partial sums at the common period $N = \text{lcm}(\text{ord } z_1, \text{ord } z_2)$ (so every oscillating factor realigns to 1 at the sample points), and accelerates the resulting smooth, power-law-convergent sequence with **Wynn’s ρ -algorithm**—the algebraic-convergence counterpart of the ε -algorithm. The ρ -algorithm yields roughly one correct digit per sample point; the evaluator inflates its internal precision to absorb the conditioning loss and returns the requested number of correct digits (`mp1_w1.py`).

4 Validation

The evaluator is checked by three independent, Wolfram-free routes plus one Wolfram cross-check.

1. **Exact symmetric closed form** (single polylogs only, an entirely separate code path):

$$\text{Li}_{a,b}(\rho^2, \rho) + \text{Li}_{b,a}(\rho, \rho^2) = \text{Li}_a(\rho^2) \text{Li}_b(\rho) - \zeta(a+b), \quad (3)$$

and the Gaussian analogue $\text{Li}_{a,b}(i, -i) + \text{Li}_{b,a}(-i, i) = \text{Li}_a(i) \text{Li}_b(-i) - \zeta(a+b)$. (The collision term is $\text{Li}_{a+b}(z_1 z_2) = \text{Li}_{a+b}(1) = \zeta(a+b)$.)

2. **Conjugation** $\text{Li}_{a,b}(\rho, \rho^2) = \overline{\text{Li}_{a,b}(\rho^2, \rho)}$.

3. **Two independent summation methods**—the ρ -accelerated cumulative sum and the Euler–Maclaurin smooth-correction series—agree to full working precision.
4. **Wolfram 15 MultiplePolyLog** at moderate precision.

All four agree; the symmetric closed form (3) holds to the full generated precision ($\sim 10^{-500}$) for every (a, b) .

5 Closed forms for the Eisenstein mixed-point values

With 500-digit values in hand, an integer-relation (PSLQ/LLL) search against the level-3 basis—monomials in the single L -values $\{\pi, \log 3, \zeta(3), \zeta(5), \text{Cl}_2(\frac{2\pi}{3}), \text{Cl}_4(\frac{2\pi}{3}), \text{Cl}_6(\frac{2\pi}{3})\}$ times the $(\rho, 1)$ generators $g_{a,b} = \text{Im Li}_{a,b}(\rho, 1)$, $h_{a,b} = \text{Re Li}_{a,b}(\rho, 1)$, and the lone (ρ, ρ) double-shuffle survivor $\text{Im Li}_{3,2}(\rho, \rho)$ —**reduces every mixed value through weight 4, and every weight-5 imaginary part**, to explicit closed forms (each canary-clean, height ≤ 10 , re-verified to 10^{-425} against the *complete* weight-graded basis). The two simplest:

$$\text{Im Li}_{1,1}(\rho^2, \rho) = -(\text{Cl}_2(\frac{2\pi}{3}) + \text{Im Li}_{1,1}(\rho, 1)), \quad \text{Re Li}_{1,1}(\rho^2, \rho) = 5 \text{Re Li}_{1,1}(\rho, 1) - \frac{1}{2} \log^2 3. \quad (4)$$

(The full list of ~ 16 reductions is in `harvest_eis_mixed.py`.) The weight-5 *real* parts and all of weight 6 do *not* reduce to this basis at height $\lesssim 10^4$.

6 The completed level-3 double-shuffle dimension

Two computations pin the dimension of the full level-3 depth-2 space—all four colored points $(\rho, \rho), (\rho^2, \rho^2), (\rho^2, \rho), (\rho, \rho^2)$ —modulo single polylogarithms and the $(\rho, 1)$ generators.

- The **exact-rank convergent double shuffle** (stuffle and shuffle of products $\text{Li}_p(s) \text{Li}_q(t)$, $s, t \in \{\rho, \rho^2\}$, together with the convergent zeta-products $\text{Li}_p(s) \zeta(q)$ whose shuffle reaches the mixed points) gives an *upper bound* of **7** through weight 6 (the divergent Li_1 words, needing shuffle regularization, are omitted).
- **Combining** that exact rank with the PSLQ closed forms of Section 5 collapses it:

weight	irreducible dimension	survivors
2	0	—
3	0	—
4	0	—
5	2	$\text{Im Li}_{3,2}(\rho, \rho)$ (<i>article</i>), $\text{Re Li}_{4,1}(\rho^2, \rho)$ (new)
6	1	$\text{Im Li}_{5,1}(\rho^2, \rho)$ (new)

So the full level-3 depth-2 space, mod single polylogs and the $(\rho, 1)$ generators, has dimension **3** through weight 6. Everything through weight 4 collapses entirely into the $(\rho, 1)$ -generator world; the mixed cube-root points contribute **exactly two new constants** beyond the article’s single (ρ, ρ) survivor—a *real* direction $\text{Re Li}_{4,1}(\rho^2, \rho)$ at weight 5 and an *imaginary* direction $\text{Im Li}_{5,1}(\rho^2, \rho)$ at weight 6.

Remark 1 (the article’s own lesson). Non-reduction by integer-relation search is strong evidence, not proof: a genuine reduction of large height is invisible to PSLQ, and the convergent double shuffle omits the regularized Li_1 relations. So the two new survivors are best read as *candidate* new level-3 depth-2 constants; a full regularized double shuffle, or a Bloch-group / Zagier-conjecture argument, would settle their independence definitively. As an additional check, an *enriched*-basis probe (`probe_eis_candidates.py`)—adding Li_n at candidate level-3 CM half-points $\frac{1+\rho}{2}$, $\frac{1}{1-\rho}$, $\frac{\rho}{\rho-1}$, $1-\rho$ (the level-3 analogue of the Gaussian $\frac{1+i}{2}$ that resolved the article’s Family-2)—fails to crack either candidate: $\text{Re Li}_{4,1}(\rho^2, \rho)$ resists a 105-atom basis (spurious scale $\sim 10^4$).

7 The Gaussian $(i, -i)$ point and an exact parallel

The same evaluator handles the level-4 Gaussian point $(i, -i)$ that the article’s level-4 run also skipped ($w = i \cdot (-i) = 1$). The picture is identical in shape.

- **Closed forms.** PSLQ against the level-4 basis (single L -values $\pi, \log 2, \beta(2), \beta(4), \beta(6), \zeta(3), \zeta(5)$, the half-point values $\text{Re Li}_n(\frac{1+i}{2})$, $\text{Im Li}_n(\frac{1+i}{2})$, and the $(i, 1)$ generators) reduces every $(i, -i)$ value through weight 3 and every weight-4 imaginary part. The weight-2 case is especially clean:

$$\text{Li}_{1,1}(i, -i) = -\text{Li}_2\left(\frac{1-i}{2}\right) \quad (\text{verified to } 10^{-50}). \quad (5)$$

- **Dimension.** The corrected exact-rank double shuffle (now including $(i, -i)$) gives an upper bound of 12; combined with the closed forms it collapses to the article’s five $(i, -1)$ constants plus **the same two new mixed directions**— $\text{Re Li}_{4,1}(i, -i)$ at weight 5 and $\text{Im Li}_{5,1}(i, -i)$ at weight 6. (One weight-3 residual, $\text{Im Li}_{2,1}(i, -1)$, is a value the article already reduces to the half-point λ/μ basis, outside the convergent double shuffle’s reach.)

Theorem 2 (the parallel). *In both towers the $w = 1$ mixed point contributes, beyond the published survivors, exactly two new depth-2 constants, and in the same two positions:*

$\text{Re Li}_{4,1}(\text{mixed})$ at weight 5, $\text{Im Li}_{5,1}(\text{mixed})$ at weight 6.

That the level-3 and level-4 mixed points add new content at identical weights and index patterns—a *real* $\text{Li}_{4,1}$ direction and an *imaginary* $\text{Li}_{5,1}$ direction—is the report’s most suggestive finding, and a natural target for the regularized double shuffle to explain.

8 Significance

- **The “one remaining piece” is closed.** The mixed cube-root points the article could not evaluate are now computable to arbitrary precision; the level-3 frontier is complete through weight 6, and the same evaluator handles the Gaussian $(i, -i)$ point the level-4 run skipped.
- **A reusable tool.** The smooth-correction + period-aware ρ -acceleration evaluator handles *every* root-of-unity double polylogarithm, including the degenerate $w = 1$ diagonal where Wolfram’s own `MultiplePolyLog` is slow and erratic (fast at low weight, stalling at higher weight). It is the natural companion to the article’s accelerated evaluator for the oscillating case.

- **A clean structural picture.** Through weight 4 the mixed cube-root doubles carry *no* new content; the first genuinely-new mixed constant appears at weight 5. This mirrors—and sharpens—the article’s finding that the colored towers are far smaller than a precision-limited search suggests.

Reproducibility

All scripts are under `src/Hypergeometric/tools/scratch/depth3-gaussian/`: `mpl_w1.py` (evaluator), `gen_eis_mixed.py` / `gen_gauss_mixed.py` (value stores `vals_eis_mixed.py` / `vals_gauss_mixed.py`), `harvest_eis_mixed.py` / `harvest_gauss_mixed.py` (closed forms), `shuffle_ds_eis_full.py` / `shuffle_ds_gauss_full.py` (exact-rank double shuffle), and `combined_dim_eis.py` / `combined_dim_gauss.py` (the combined dimension). Values are generated to 500 digits; every displayed identity is verified to at least 10^{-425} , and the symmetric closed form (3) (an independent single-polylog identity) holds to $\sim 10^{-500}$ for every value. The verified closed forms and the dimension results are collected in the `Get[]`-able store `src/PolyLog/identities/discovered-mixed-cuberoot-doubles.wl`.